Road-supported effective constitutive coupling and anisotropic routing responses in urban mobility

Tiehang Xu[1] Jiahui Wang[2] Junyan Yang[3]
Qiao Wang[1,2,*]

[1]School of Information Science and Engineering, Southeast University, Nanjing, 210096, China
[2]School of Economics and Management, Southeast University, Nanjing, 210096, China
[3]School of Architecture, Southeast University, Nanjing, 210096, China
Emails: tiehangxu@seu.edu.cn; wangjh0521@seu.edu.cn; yjy-2@163.com; qiaowang@seu.edu.cn
∗Correspondence: qiaowang@seu.edu.cn

Abstract

Urban movement is not determined solely by where journeys begin and end. Here we identify an effective constitutive relation linking the source–sink balance of observed flows to anisotropic routing. Using a Hodge decomposition, we estimate this relation as a linear operator within a divergence-free space supported by independently registered roads. Across Porto, San Francisco and Chengdu, held-out departure scores indicate a partial and city-dependent source–routing association, strongest in Porto, weaker in San Francisco and small in Chengdu. The fitted response operator reveals strong local anisotropy. The road-defined support is specific against rank-matched random subspaces, but in the tested matched-capacity comparison a road-only medium is not sufficient and does not by itself explain the coupling. These results characterize a partial, observation-dependent effective linear coupling rather than a general constitutive law of cities.

Cities organize movement as well as activities. The location of homes, jobs and services determines where journeys begin and end, while the arrangement of streets shapes the spaces through which those journeys pass. Neighbourhoods with similar net inflows can nevertheless experience different through-movement, and a redistribution of activity need not produce a proportional redistribution of traffic. Gravity, radiation and spatial-interaction models describe flows between places [1, 2, 3, 4, 5]; traffic assignment links demand to routes through travel costs and behavioural rules [6, 7]; and graph-based learning offers flexible descriptions of traffic and assignment [8, 9]. Between aggregate interaction and individual routes lies a useful urban-scale object: the local direction and intensity of collective movement.

Field descriptions make this intermediate scale accessible. Recurrent commuting can be represented by a vector field whose potential identifies mobility basins [10]; trajectory-based formulations extend the approach beyond commuting [11]; and Laplacian evolution identifies urban source–sink patterns [12]. Hodge decomposition separates a potential-driven component, associated with the imbalance between arrivals and departures, from circulation that redistributes movement without changing that balance [13, 14]. The two sectors are geometrically orthogonal, but spatial orthogonality does not preclude reproducible statistical coupling. Related field studies identify scaling, temporal response, directional organization and anomalous movement [15, 16, 17, 18]. Whether changes in one sector systematically accompany changes in the other is the question that turns a descriptive decomposition into an empirical relation.

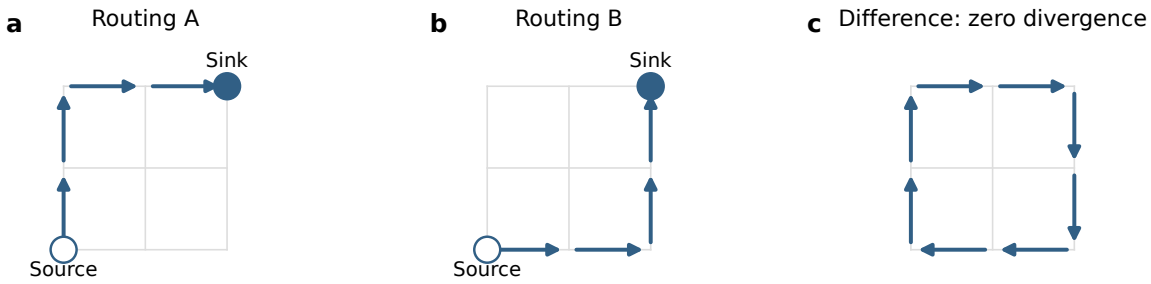


Figure 1: Routing freedom under conservation. a,b, Two routes share a source and sink. **c**, Their difference has zero divergence. Balance fixes the source–sink pattern but leaves routing undetermined.

We approach this question through an effective constitutive relation: a response connecting the source–sink organization of observed flows to the routing component left undetermined by balance. Roads provide a spatial medium that can be measured independently of the mobility values used to estimate the response [19, 20, 21, 22, 23]. The resulting operator is an effective, data-estimated description, not a claim that road geometry determines flow. It asks whether a compact linear mapping can reproduce part of the routing organization, and whether that mapping is specific to the road-defined support rather than to any conservation-compatible subspace of the same size.

Here we estimate this response from trajectories in Porto, San Francisco and Chengdu. UrbanFlow constructs a road-supported routing space, fits a linear source–routing operator on training dates and evaluates it on held-out dates. We examine predictive content, frequency-resolved response, directional anisotropy and the limits of the road medium. The central finding is a measurable but partial and city-dependent effective linear coupling; its strength is strongest in Porto and weaker in San Francisco and Chengdu, where recurrent structure already explains much of the observed routing field.

1 Results

1.1 Formulation of the effective response

We represent hourly movement as signed crossings of the faces between spatial cells. For a flux field F_t, its discrete divergence $x_t = DF_t$ records the cellwise source–sink balance. Here x_t denotes the divergence of the observed coarse-grained flux, not an independently measured origin–destination demand field. The field separates into a minimum-norm component fixed by that balance and a divergence-free routing component,

$$F_t = D^{\dagger} x_t + F_t^{\circ}, \qquad DF_t^{\circ} = 0. \tag{1}$$

Figure 1 illustrates the physical freedom: different directional configurations can carry the same source and sink. The two components are spatially orthogonal, but their organization can covary across observations. Predicting the routing component from the source therefore tests a statistical relationship between the irrotational and divergence-free sectors of the measured flow.

Road geometry defines an orthonormal basis U_R within the divergence-free space. The road-coordinate state is $y_t = U_R^{\top} F_t^{\circ}$; a fitted response K_R relates this state to the source. The corre-

sponding operator in physical space is

$$\widehat{F}_t^{\circ} = C_R x_t, \qquad C_R = U_R K_R P_x, \tag{2}$$

where P_x restricts inputs to conservation-compatible source patterns. Roads supply the structural support; dated mobility observations determine the response within it. This separation makes the operator interpretable as an effective property of movement in a given urban medium.

The road space captures 0.904, 0.886 and 0.731 of the observed routing energy in the P/D-aligned observation pools for Porto, San Francisco and Chengdu, respectively (Supplementary Table S2). Three claims must be kept separate: road support represents much of the routing field; the road-induced subspace is specific against rank-matched random subspaces; and the source predicts routing within that support. The first two are tested below and the third by the P/D scores. None of them establishes that road geometry determines the coupling.

1.2 Core result: constitutive coupling across three cities

Operators are estimated on training dates and evaluated on dates excluded from fitting and model selection. Two evaluations distinguish reproduction of the observed field from prediction of its day-specific variation: P scores the raw field, whereas D removes the training-estimated recurrent hourly pattern from both source and target (Methods). Figure 2a–c and Supplementary Table S2 report both on a common pooled-error convention (Supplementary Note S6).

The raw ROAD scores are 0.682 in Porto, 0.597 in San Francisco and 0.606 in Chengdu, whereas the corresponding source-free climatologies score 0.472, 0.727 and 0.931 (diamonds in Fig. 2a–c). Raw-field reproduction therefore cannot by itself be read as source-specific information. Prediction of departures remains positive in all three cities, with ROAD scores of 0.468, 0.240 and 0.120. Paired-date bootstrap intervals are 0.355–0.528 in Porto, 0.188–0.288 in San Francisco and 0.107–0.142 in Chengdu (descriptive, three dates). A source–routing pairing null that permutes date blocks within TRAIN, VAL and TEST and refits the operator yields scores near zero; no Porto or San Francisco null draw reaches the observed value (empirical $p = 1/200$ with 199 permutations). Chengdu's three dates allow only six date permutations, so its null is descriptive rather than a significance test. Adding the D departures to the TRAIN climatology gives a full-field gain of $\Delta R_I = (1 - R_{\mathrm{CLIM}})R_D \approx 0.25$, 0.07 and 0.008 in Porto, San Francisco and Chengdu. The evidence is therefore hierarchical: Porto shows the clearest conditional relation, San Francisco a weaker but consistent one, and Chengdu a highly recurrent field in which source information adds little beyond the daily pattern. Chengdu also has only three TEST dates in this protocol, so its departure score locates the scale of the association rather than supporting a strong cross-date claim. Representative physical reconstructions for all three cities are retained in Supplementary Fig. S6.

The three cities also differ in the spatial concentration of routing (Fig. 2d–f). A single TRAIN-derived spatial mode captures 0.433 of held-out routing energy in Porto, 0.468 in San Francisco (fold mean) and 0.810 in Chengdu; eight modes capture 0.616, 0.853 and 0.946. Chengdu thus exhibits a compact, reproducible spatial organization. This descriptive concentration and the source-based prediction above characterize different properties of the same field: a compact set of observed patterns does not by itself determine their response to changing sources.

The road restriction is informative but incomplete. Full-face ridge and a training-only POD representation equal or outperform it on the same target (Supplementary Table S2); D/FACE ridge scores are 0.504, 0.262 and 0.126. Roads therefore provide an independently specified, interpretable medium rather than the highest attainable linear reconstruction accuracy.

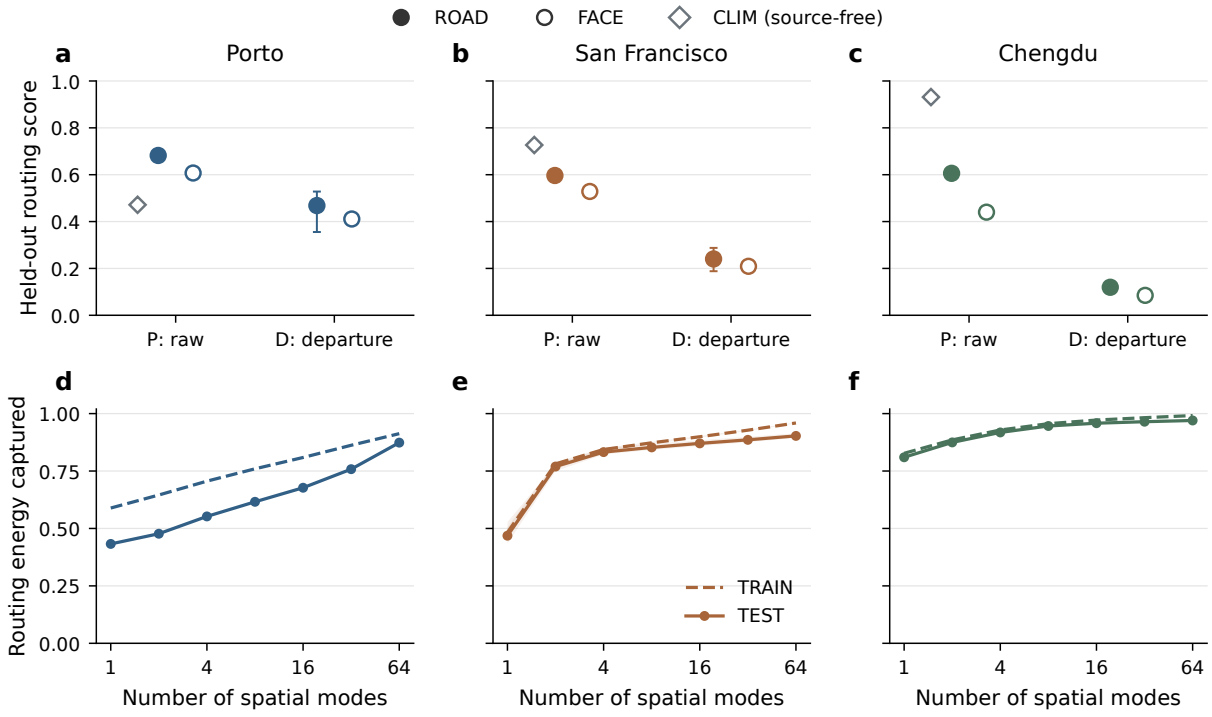


Figure 2: Coupling and spatial organization. a–c, P/D prediction scores on ROAD (filled) and FACE (open) support, pooling TEST errors; diamonds show the source-free P/ROAD climatology. D/ROAD points carry 95% paired-date bootstrap intervals (Chengdu's interval is descriptive). **d–f**, Energy captured by TRAIN-derived spatial POD modes on TRAIN (dashed) and TEST (solid); mode count is logarithmic. San Francisco curves show fold means and ranges. Chengdu uses the 14-date, 06–23 h window. POD capture measures representation, not source-based prediction. The distance between the P point and the climatology diamond is the information available to the source; in Chengdu the P score lies below the climatology.

1.3 Frequency-domain check

The descriptive joint spectra are reported in Supplementary Fig. S1. They show how temporal rhythms are distributed across spatial modes but do not by themselves test the constitutive relation. Chengdu's time-independent component carries 0.668 of source energy and 0.811 of full-face routing energy; its 18-point transform resolves the observed 06–23 h block, whereas Porto and San Francisco use complete 24-hour days. These spectra are descriptive: a compact or persistent pattern does not imply that the source predicts its day-specific routing.

The frozen operator is instead evaluated frequency by frequency without fitting a new temporal response (Fig. 3). Because the prediction and observation are complex Fourier coefficients, we report a phase-sensitive score,

$$R_m^2 = 1 - \frac{\sum_d \|a_{d,m} - b_{d,m}\|^2}{\sum_d \|a_{d,m}\|^2}, \tag{3}$$

and the pooled signed alignment

$$\widetilde{\gamma}_m = \frac{\sum_d \mathrm{Re}(a_{d,m}^* b_{d,m})}{\sqrt{\sum_d \|a_{d,m}\|^2}\sqrt{\sum_d \|b_{d,m}\|^2}}, \tag{4}$$

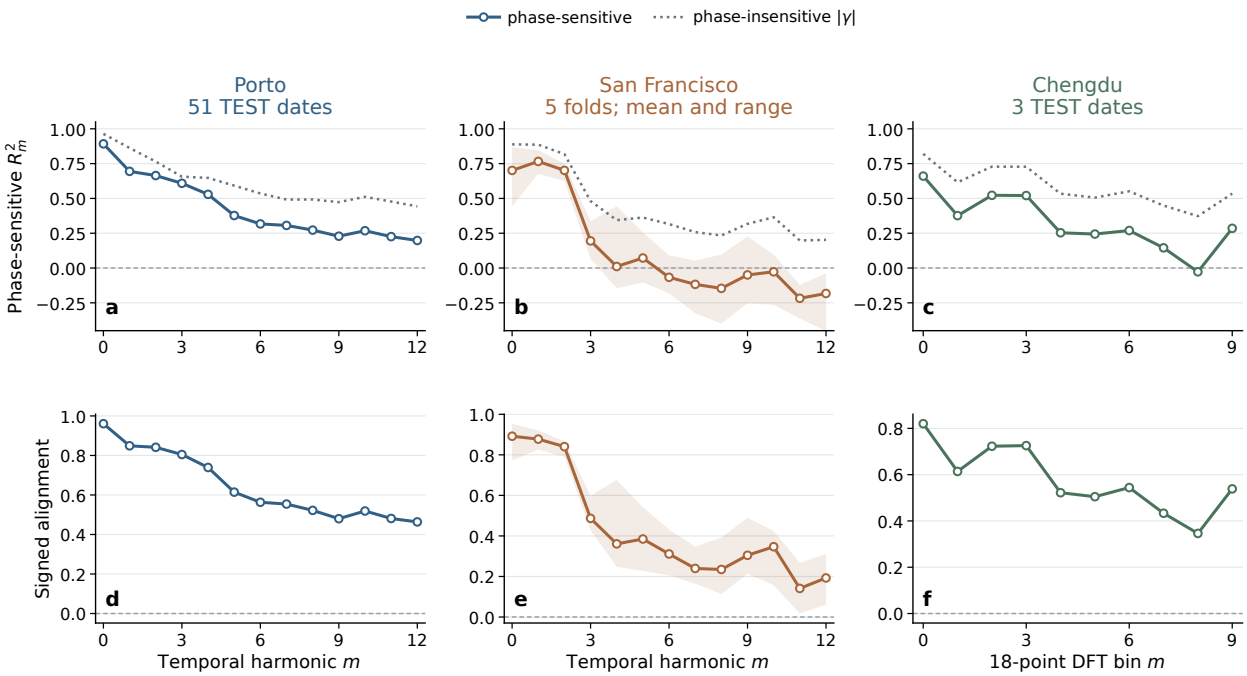


Figure 3: Phase-sensitive frequency-resolved response. **a–c**, Per-bin R^2_m using per-date complex coefficients. **d–f**, Signed alignment $\widetilde{\gamma}_m$, which retains phase and direction. The dotted curve in **a–c** is the phase-insensitive magnitude $|\gamma_m|$ used previously. The frozen P-selected operator is used throughout. San Francisco shows fold means and ranges. Porto and San Francisco use 24-point DFTs; Chengdu uses all ten single-sided bins of its 18-point observed-block DFT.

where $a_{d,m}$ and $b_{d,m}$ are observed and predicted coefficients on date d (Methods). The phase-insensitive magnitude $|\gamma_m|$ used previously is shown for comparison. In Porto, R^2_m falls from 0.89 at $m = 0$ to about 0.20 at the highest harmonics while $|\gamma_m|$ remains near 0.46 and the pooled signed alignment remains 0.46–0.96. San Francisco and Chengdu have several bins with $R^2_m < 0$; Chengdu's R^2_m is 0.66 at $m = 0$ and 0.38 at $m = 1$. The high phase-insensitive alignment reported previously therefore does not establish a correct complex response; the phase-sensitive scores are the relevant test.

1.4 Second result: directional organization and the limits of the road medium

Response anisotropy exceeds road-support anisotropy in all three cities (Fig. 4a–c; full distributions in Supplementary Fig. S4). Median response values are 0.635, 0.651 and 0.705 in Porto, San Francisco and Chengdu, against structural-tensor values 0.144, 0.057 and 0.280. San Francisco's map is stable across date folds (mean pairwise spatial correlation 0.984). Strong local anisotropy persists under source-covariance weighting (median 0.77, 0.93 and 0.76), indicating that directional selectivity is also pronounced under the realized excitation statistics. The orientation of these tensors is a separate question. The archived joint-anisotropy-weighted axis agreement with road support is 0.79, 0.79 and 0.76. A spectrum-matched null that rotates K_R within the road output space while preserving its spectrum yields null agreement of 0.97, 0.91 and 0.81 (median of 999 draws); all 999 null draws exceed the archived estimate in every city (empirical $p = 1.0$). Axis agreement is therefore not distinguishable from the spectrum-matched null and provides no independent evidence of

a road-aligned response. We retain the anisotropy magnitude as an operator property but do not interpret its orientation as independent validation.

Road capture is 0.90, 0.89 and 0.73 in the P/D-aligned pools. The rank-matched null is evaluated on the full aligned observation pool, which is distinct from the P/D-aligned pool used for the held-out scores. Against 999 rank-matched random subspaces of $\ker D$, every random basis leaves more energy uncaptured than the road-induced basis (Fig. 4d). Excess ΔE spans 0.23–0.30, 0.28–0.33 and 0.09–0.15 across the cities. The ambient space and rank are fixed, but each random basis spans a different subspace. This test concerns representation specificity against that null; it does not compare all structured alternatives, such as rotated or rewired road networks, and does not show that roads determine the response.

The matched-capacity test asks whether the road medium is sufficient (Fig. 4e). At $R = n_{\mathrm{int}}$ (1984, 5512 and 7080 nominal pre-normalization block values), road gains over the constant medium are 0.009, 0.010 and -0.001, compared with mobility-inferred gains of 0.092, 0.085 and 0.036. The paired mobility-minus-road differences are 0.083 (95% interval 0.060–0.109), 0.075 (0.062–0.091) and 0.038 (0.035–0.040). Permuting the mobility medium removes its gain (-0.047, -0.052 and -0.016). Road patterns come from an external GIS raster, whereas mobility patterns are learned from TRAIN observations; equal R matches nominal capacity and estimator, not information. The score uses the raw routing target with each TEST date's own mean as reference, not D's TRAIN-phase departure target. Chengdu's four alternate-hour dates make its interval a small-sample descriptive summary. Thus the road-induced subspace is specific under the stated null, while the road-only medium is not sufficient under this estimator and information regime (Supplementary Note S9).

1.5 Comparison with mobility modelling approaches

Table 1 separates source-to-routing inference, OD estimation and network-loading diagnostics. The inference block uses archived per-date errors pooled within each city for every row, so the rows share an aggregation convention. The San Francisco Graph WaveNet cell is pooled from archived per-date numerators and denominators (0.254), not averaged from fold R^2 values. Loading rows use different information, including TRAIN margins or exact TEST OD, and diagnose the implemented pipelines rather than modelling paradigms.

On the inference task, full-face ridge and POD exceed the road-restricted operator, while UrbanFlow exceeds the completed Graph WaveNet adaptation under the common pooled convention: 0.607 versus 0.497 in Porto, 0.528 versus 0.254 in San Francisco and 0.440 versus 0.275 in Chengdu. The archived Porto paired contrast (Graph WaveNet minus UrbanFlow) is -0.1102 (95% interval $[-0.1762, -0.0268]$). Sealed-seed robustness for Porto and Chengdu, and the archived per-date contrast, are reported in Supplementary Note S5. Sealed San Francisco per-date errors were not archived, so the sealed fold-mean value is not used for ranking. Sealed reruns confirm that UrbanFlow leads at every realised seed in Porto and Chengdu by at least 0.019 and 0.090; a sealed pooled comparison is not available for San Francisco because its per-date errors were not archived. These results concern the registered 300-epoch adaptation and its validation rule, not neural models generally.

Gravity attains native OD-overlap scores of 0.525, 0.464 and 0.555, but its loading rows and exact-TEST-OD AON rows are not deployable source-only predictions: they use TEST-fitted amplitudes or additional TEST demand. Chengdu has the highest gravity CPC yet near-zero loading scores. Destination permutation leaves its loaded field almost unchanged (cosine 0.93–0.94), indicating a largely margin-driven pattern that does not match observations (cosine 0.04). Rebuilding

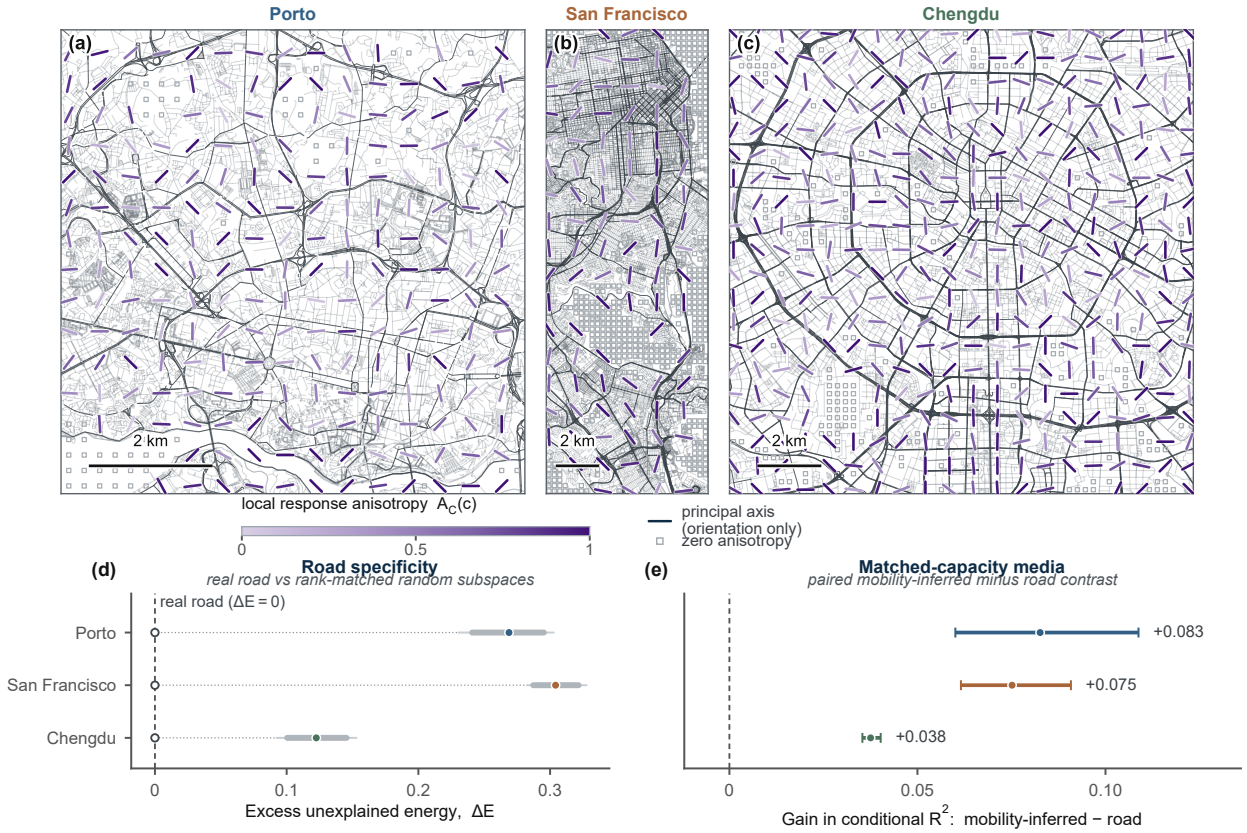


Figure 4: Road specificity without constitutive sufficiency. a–c, Response anisotropy A_C on complete 250 m regular interior lattices at true aspect (San Francisco: five-fold consensus). Colour shares 0–1; segments show axial directions for $A_C > 0.02$ and hollow squares mark zero anisotropy. Streets are © OpenStreetMap contributors. **d**, ΔE for 999 rank-matched random subspaces in fixed $\ker D$. Thick/thin marks show 1–99%/full null ranges, not confidence intervals; 0/999 nulls are at least as good as roads, empirical $p = 1/1000$. This null tests representation specificity, not all structured road alternatives. **e**, Paired mobility-minus-road ΔR^2 at $R = n_{\text{int}}$ with 95% paired-date intervals. Scores use raw routing against each TEST-date mean, not P or D. GIS roads and TRAIN-derived mobility match capacity, not information. Chengdu has four alternate-hour TEST dates, so its interval is descriptive. Axis agreement with road support is not in excess of a spectrum-matched rotation null and is therefore not interpreted as independent validation (Results; Supplementary Note S8). Media: Supplementary Fig. S5.

from occupied runs alone retains 84% of absolute flux and gives 0.929 cosine with the all-fix field, so this population mismatch alone is insufficient to explain the loading failure. These rows diagnose the implemented demand–registration–loading pipeline, not traffic assignment as a paradigm. Radiation, user equilibrium and Deep Gravity were not run; Graphormer's reduced-budget result is excluded. Detailed protocols are in Supplementary Notes S5 and S10.

Table 1: Mobility model comparisons.

Model	Metric	Porto	San Francisco	Chengdu
Source-to-routing inference				
UrbanFlow	$R^2_{\mathrm{P,FACE}}$	0.607	0.528	0.440
Full-face ridge	$R^2_{\mathrm{P,FACE}}$	0.701	0.611	0.572
POD regression	$R^2_{\mathrm{P,FACE}}$	0.701	0.637	0.575
Graph WaveNet	$R^2_{\mathrm{P,FACE}}$	0.497	0.254	0.275
OD estimation				
Gravity	CPC	0.525	0.464	0.555
Network-loading diagnostics				
Gravity + AON	R^2_{FACE} [a]	0.136	0.270	< 0.001
TEST OD + AON	R^2_{FACE} [a]	0.172	0.336	0.002
TEST OD + AON	R^2_{FACE} [b]	0.126	-0.533	-0.565

Inference rows share sources, splits and targets; TEST errors are pooled from per-date numerators and denominators for every row, including Graph WaveNet. Gravity uses TRAIN OD margins; AON denotes all-or-nothing assignment. [a]Amplitude fitted on TEST. [b]Fixed amplitude, $\alpha = 1$. Graph WaveNet reports archived single draws; sealed seed means and the archived Porto paired contrast are in Supplementary Note S5. Sealed San Francisco per-date errors are not archived, so its fold-mean value is not used for ranking. CPC and loading are separate tasks. Chengdu loading is margin-dominated (destination-permutation cosine 0.93–0.94), so its loading cells do not rank demand models. Full protocols: Methods and Supplementary Notes S5 and S10.

2 Discussion

2.1 A constitutive description of urban movement

The identified operator connects two aspects of urban mobility that conservation alone leaves unrelated: source–sink organization and divergence-free routing. Hodge-based descriptions expose potential landscapes and circulation [13, 14]; the constitutive step measures how these sectors covary across observations. Spatial orthogonality at one instant does not imply statistical independence across dates. A reproducible map between the sectors therefore adds empirical information to the decomposition, rather than restating balance. The relation is partial and city-dependent: it is strongest in Porto, weaker in San Francisco and adds little full-field information in Chengdu. Because source and target are constructed from the same trajectories, this is a held-out conditional reconstruction, not a forecast from independently observed demand.

The physical object is the lifted operator C_R, which maps a distributed source pattern to a directional routing response. Its columns describe model-implied responses to compatible source directions, and its local tensors describe directional selectivity. These properties turn a prediction matrix into a description of urban organization at the observation scale. Phase-sensitive frequency scores and local anisotropy then measure different aspects of the same operator without equating predictive accuracy with a complete account of the city.

2.2 Urban anisotropy and the role of roads

Anisotropy distinguishes the geometry available to movement from the directional response expressed through it. Strong local selectivity varies across space rather than defining one city-wide axis, and its magnitude persists when the response is weighted by the TRAIN source covariance. The orientation of these tensors is less informative: axis agreement with road support does not

exceed a spectrum-matched rotation null. We therefore treat the tensors as a property of the fitted operator on its identified source span, not as independent evidence that sampled activity reweights road geometry in a road-aligned way. This connects street configuration [24, 25, 26] to a measured operator while keeping the interpretation conditional.

Roads act as a testable structural hypothesis, but only within a limited claim. The rank-matched null shows that the road-induced subspace is specific against that null; it does not compare rotated, rewired or otherwise structured road networks, and it does not show that the source predicts routing because of road geometry. The matched-capacity and unrestricted controls show that a road-only medium is not sufficient. The appropriate conclusion is a road-supported effective coupling, not a road-determined constitutive law. Independent geometry makes the support interpretable; it does not make the estimated response independent of activity, scale or observation regime.

2.3 Scope and further tests

The three cities establish a measurable but partial source–routing relation. Their different departure scores quantify how much day-specific organization is resolved by the selected input; substantial residuals remain. Signed source balance compresses movement: different OD pairings and opposing flows can produce the same net balance. Sampling and excitation also limit identification, particularly for Chengdu's short date window and San Francisco's restricted source span. These mechanisms cannot be separated by the present comparisons. A decisive test of the road hypothesis would compare the observed road support with rotated or rewired road networks at the same rank, input and target; the rank-matched random-subspace null used here does not substitute for that structural intervention. Additional state variables, nonlinear responses and more general memory remain testable extensions.

The operator is an effective description of the sampled city, not an intervention model. Sources and targets come from the same trajectories, and each city's operator is fitted separately. Forecasting from independently available demand, transfer between cities and responses to infrastructure changes require further validation. The present contribution is a partial, identifiable effective object: a relation whose structural support, directional organization and unexplained variation can be measured and challenged separately, without equating a road map, conservation or predictive accuracy with a general law of urban movement.

3 Methods

3.1 Data and discrete fields

Porto uses ECML/PKDD taxi trajectories [27]: 252 dates, a chronological TRAIN/VAL/TEST split of 151/50/51 dates and a 32×32 lattice. San Francisco uses Cabspotting [28]: 15 dates, five blocked folds of 9/3/3 dates and a 32×88 lattice. Chengdu uses DiDi ride-hailing trajectories [29] with 30-second sampled fixes and a 60×60 lattice. Its P/D, coherent-response, anisotropy and spectral analyses use a 14-date 8/3/3 subset from the continuous-observation regime. The matched-capacity medium sweep uses a separate chronological 10/4/4 split whose four held-out dates are 26–29 August 2014; these dates belong to the alternate-hour delivery phase, with nine alternating recorded hours per date. The structural null is evaluated on the 19 valid dates of the aligned grid. These protocols are related but not interchangeable. The hourly phase sets are 00–23 h for Porto/San Francisco and 06–23 h for Chengdu. Each city is estimated separately. Observation regimes and frozen date lists are specified in Supplementary Note S1 and Supplementary Table S1.

Successive trajectory segments are walked through the aligned nominal 250 m cells. Signed crossings of each interior face are accumulated in 60-minute windows to form $F_t \in \mathbb{R}^{n_f}$. No map matching is applied. The source x_t is the divergence of this observed crossing field, not an independently measured origin–destination demand field; it can also reflect changes in vehicle stock, boundary crossings and observation support. For the cell–face incidence matrix $D \in \mathbb{R}^{n_c \times n_f}$,

$$x_t = DF_t, \quad P_0 = I - D^\dagger D, \quad F_t^\circ = P_0 F_t, \quad P_x = DD^\dagger. \tag{5}$$

The routing target is calculated from the same crossing field as the source. The Laplacian solve is gauge-pinned.

The archived footprint mask is the union of cells with non-zero density over all dates, (N > 0).any(0). It is therefore transductive. It is used only as a fixed analysis and display footprint; it is not an input to D, U_R or K_R, which depend on the grid, study domain and road registration. A TRAIN-only footprint is a strict subset of this mask (987 versus 994 cells in Porto, 2345 versus 2532 in one San Francisco fold), but the core operator and scores do not change with that substitution because the mask does not enter their construction.

3.2 Road space and response estimation

The drivable OpenStreetMap layer [30] is registered to the study complex. Let B_R be its node–edge incidence, A_R the edge-to-face lift and Π the registration map satisfying $DA_R = \Pi B_R$. A cycle basis Z_R satisfies $B_R Z_R = 0$. The Gram eigendecomposition of $A_R Z_R$ gives

$$U_R = \mathrm{orth}(A_R Z_R), \qquad U_R^\top U_R = I, \qquad DU_R = 0, \qquad y_t = U_R^\top F_t^\circ. \tag{6}$$

Geometry, study domain and road registration determine U_R; the activity footprint mask and mobility flux values are not inputs. Supplementary Note S2 gives the registration conventions.

For any source or target v_t, TRAIN phase means and departures are

$$\bar{v}_\phi = \frac{1}{n_{\mathrm{tr},\phi}} \sum_{t \in \mathrm{TRAIN}:\phi(t)=\phi} v_t, \qquad \delta v_t = v_t - \bar{v}_{\phi(t)}. \tag{7}$$

Stack $\delta x_t^\top$ and $\delta y_t^\top$ as the rows of X and Z. Ridge estimation yields

$$\begin{aligned} K_R^\star &= \arg\min_K \{\|Z - XK^\top\|_F^2 + \lambda \|K\|_F^2\} \\ &= Z^\top X (X^\top X + \lambda I)^{-1}, \qquad C_R = U_R K_R^\star P_x. \end{aligned} \tag{8}$$

We use $\lambda = \kappa s_{\max}(XX^\top)$ and select κ on VAL from the grid specified in Supplementary Note S2. The archived structural-characterization estimators use $\kappa = 0.01$ in Porto, 0.1 in San Francisco and 0.1 in Chengdu's D frame; the archived Chengdu P-frame sensitivity uses $\kappa = 0.001$. Comparison arms select their own regularization in the specified scoring frame. Full-face ridge fits F_t° and projects predictions with P_0. POD estimates its output basis from TRAIN departures and selects rank and regularization using FACE validation error. Since $K_R P_x = K_R$, C_R and K_R have the same nonzero singular values.

3.3 Scoring

Define $T_t = U_R^\top F_t^\circ$ on ROAD support and $T_t = F_t^\circ$ on FACE support. Let B_{P} and B_{D} denote the fitted maps on that support. Both use TRAIN departures; P selects on raw VAL inputs/targets, while D selects on VAL departures using TRAIN means. Predictions have no fitted intercept:

$$\begin{aligned} R^2_{\mathrm{P}} &= 1 - \frac{\sum_{t\in\mathrm{TEST}} \|T_t - B_{\mathrm{P}} x_t\|^2}{\sum_{t\in\mathrm{TEST}} \|T_t\|^2}, \\ R^2_{\mathrm{D}} &= 1 - \frac{\sum_{t\in\mathrm{TEST}} \|\delta T_t - B_{\mathrm{D}} \delta x_t\|^2}{\sum_{t\in\mathrm{TEST}} \|\delta T_t\|^2}. \end{aligned} \tag{9}$$

Numerators and denominators are pooled across TEST observations and folds within each city. The denominators are target energies, not variances about a TEST mean. The source-free reference is $\bar{T}_{\phi(t)}$, reported in Supplementary Table S2.

3.4 Structural tests

For any orthonormal routing basis U, the uncaptured-energy statistic is

$$E(U) = 1 - \frac{\sum_t \|U^\top F_t^\circ\|^2}{\sum_t \|F_t^\circ\|^2}, \qquad \Delta E_b = E(U_b) - E(U_R). \tag{10}$$

Let K_D be an orthonormal basis of $\ker D$. Each null draw uses $G_{ij} \overset{\mathrm{iid}}{\sim} N(0,1)$, $Q = \mathrm{orth}(G)$ and $U_b = K_D Q$ at rank $\mathrm{rank}(U_R)$; thus the subspace distribution does not depend on a non-orthogonal factorization of $\ker D$. The test uses the full observation pool of each city's aligned grid and involves no source-response fit; these pools differ from the frozen three-city P/D comparison.

The second test compares road, mobility-inferred and spatially permuted mobility media at equal capacity. Each per-face medium is reduced to the same nominal block capacity R on nested spatial partitions before mean-1/floor normalization; R is not the effective degrees of freedom after that normalization. $R = 1$ gives the constant-medium reference. Permutation preserves the values, partition and nominal block count. The archived statistic is the equal-date mean of

$$R^2_{\mathrm{cond},d} = 1 - \frac{\sum_{t\in d} \|T_t - \widehat{T}_t\|^2}{\sum_{t\in d} \|T_t - \langle T\rangle_d\|^2}. \tag{11}$$

Figure 4e reports the paired-date interval for mobility minus road at the largest archived capacity, $R = n_{\mathrm{int}}$, the number of interior faces (1984, 5512 and 7080); Supplementary Fig. S5 reports each medium's gain over the constant medium at that capacity. Chengdu archives a single mobility channel (all motion), so its sweep uses that channel on the frozen per-fold basis, where San Francisco's uses the occupancy-defined channel; its interval resamples four held-out dates. This date-centred statistic is separate from Eq. 9. Source files and partitions are specified in Supplementary Note S4.

3.5 Anisotropy

For each regular interior cell, L_c sums signed face flux into two physical components. Define

$$\begin{aligned} \Sigma_C(c) &= L_c C_R C_R^\top L_c^\top, \\ \Sigma_U(c) &= L_c U_R U_R^\top L_c^\top, \\ A(c) &= \begin{cases} (\lambda_1(c) - \lambda_2(c))/(\lambda_1(c) + \lambda_2(c)), & \lambda_1(c) + \lambda_2(c) > 0, \\ 0, & \lambda_1(c) + \lambda_2(c) = 0. \end{cases} \end{aligned} \tag{12}$$

Here $\lambda_1 \geq \lambda_2$ are the eigenvalues of the relevant tensor. The principal eigenvector defines an axial orientation θ modulo π. San Francisco uses the per-cell mean anisotropy and nematic mean orientation over five folds. Because Σ_C uses a unit input metric, we also evaluate the source-covariance-weighted response tensor

$$\Sigma_{\text{response}}(c) = L_c C_R \Sigma_{xx} C_R^\top L_c^\top, \tag{13}$$

with Σ_{xx} the TRAIN covariance of source departures. Structural-axis agreement is evaluated on regular interior cells for which both anisotropies are nonzero, using the frozen joint-anisotropy weighting

$$G_\theta = \frac{\sum_c A_C(c) A_U(c) \cos[2(\theta_C(c) - \theta_U(c))]}{\sum_c A_C(c) A_U(c)}. \tag{14}$$

To test whether this agreement exceeds what the shared road subspace and operator spectrum imply, write $K_R = Q\Lambda V^\top$ and replace Q by a Haar-random orthonormal frame Q_0 in the road output space, $K_0 = Q_0 \Lambda V^\top$, preserving the spectrum and input directions. We recompute G_θ for 999 draws and compare the observed value with this null.

Only regular interior cells enter the anisotropy statistics; their metric $L_c L_c^\top = 2I$ is isotropic. Metric whitening, orthogonal changes of road coordinates and rigid physical rotations are checked as in Supplementary Note S4.

3.6 Spectra and coherent response

For the joint spectrum in Supplementary Fig. S1, each TRAIN date is transformed separately, retaining its mean. For field $v_{d,j}$ and observed block length N,

$$\begin{aligned} Z_{d,m} &= N^{-1/2} \sum_{j=0}^{N-1} v_{d,j} e^{-2\pi i m j/N}, \qquad f_m = m/N, \\ P_m &= w_m \langle \|Z_{d,m}\|^2 \rangle_d, \qquad J_{m,k} = \frac{P_m}{\sum_l P_l} \frac{s_{m,k}^2}{\sum_a s_{m,a}^2}. \end{aligned} \tag{15}$$

$s_{m,k}$ are singular values of the date-by-space coefficient matrix. The weights are $w_0 = w_{N/2} = 1$ and $w_m = 2$ otherwise. Porto/San Francisco use $N = 24$; Chengdu uses $N = 18$ with no night-time imputation. Frequencies are cycles per observed hour. San Francisco averages P_m and within-frequency mode shares separately across folds before multiplication. The first eight modes are displayed after full-spectrum normalization. Spatial POD curves in Fig. 2 use a raw TRAIN routing basis and evaluate projection energy on TRAIN/TEST.

For the frequency-resolved response, each TEST date is transformed with the same unitary DFT used for the descriptive spectrum: 24 points in Porto/San Francisco and 18 points in Chengdu. Let $a_{d,m}$ and $b_{d,m}$ be the observed and predicted road-coordinate coefficients on date d at bin m. We report

$$R_m^2 = 1 - \frac{\sum_d \|a_{d,m} - b_{d,m}\|^2}{\sum_d \|a_{d,m}\|^2}, \qquad \widetilde{\gamma}_m = \frac{\sum_d \mathrm{Re}(a_{d,m}^* b_{d,m})}{\sqrt{\sum_d \|a_{d,m}\|^2} \sqrt{\sum_d \|b_{d,m}\|^2}}, \tag{16}$$

where the signed alignment pools date-wise complex inner products and energies across TEST dates. R_m^2 is phase-sensitive and detects phase or direction errors; $\widetilde{\gamma}_m$ retains phase and direction. The phase-insensitive magnitude $|\gamma_m| = |a_m^* b_m|/(\|a_m\|\|b_m\|)$ is reported only for comparison. The P-selected operator is fixed. All ten single-sided bins are evaluated in Chengdu.

3.7 Baselines and uncertainty

Graph WaveNet uses the frozen splits, a 300-epoch budget and four-point VAL-selected grid, with phase-centred TRAIN data and raw VAL/TEST data (P/FACE). Archived cells are single draws with pooled per-date errors; sealed cells repeat construction and VAL selection at each declared seed. Table 1 uses the archived pooled draws for all three cities, including a San Francisco value pooled from its archived per-date errors. Sealed seed means for Porto and Chengdu, and the archived Porto paired contrast, are reported in Supplementary Note S5; sealed San Francisco per-date errors were not archived, so its sealed fold-mean is not used for ranking. Graphormer's 60-epoch run is excluded. Configurations and outputs are indexed in Supplementary Note S10.

Gravity estimates OD flows from TRAIN margins and distance, selecting friction on VAL. Its native overlap is CPC $= 2\sum_{ij}\min(\widehat{Q}_{ij}, Q_{ij})/(\sum_{ij}\widehat{Q}_{ij} + \sum_{ij} Q_{ij})$ at 4×4-cell aggregation. AON uses time-shortest paths, road costs and a 150 m snap threshold; gravity loading retains the top 150,000 zone pairs. Scaled diagnostics fit α^* on TEST, while the fixed-amplitude arm uses $\alpha = 1$. Registration drops 27.7%/8.9%/22.3% of Porto/San Francisco/Chengdu trips. Chengdu OD uses maximal occupied runs under the frozen cleaning rules. Its observed field counts crossings within each hour, whereas loading routes trips by origin window; it also uses all fixes, whereas OD uses occupied runs only. Rebuilding from occupied runs removes 16.3% of absolute flux and gives 0.93 pooled cosine with the all-fix field. This is one reconstruction, not a universal upper bound for every loader, and does not by itself explain the loading failure. No radiation or user-equilibrium score is estimated (Supplementary Note S5).

Paired bootstrap intervals for the primary D scores use 10,000 resamples of held-out dates within each fold and the 2.5/97.5 percentiles; faces are never resampled. The source–routing pairing null permutes complete date blocks within TRAIN, VAL and TEST separately, refits the ridge operator and re-selects κ in each of 199 draws. Neural contrasts use 9,999 resamples within folds with seed 20261003 and Holm adjustment within city/support, whereas the matched-capacity mobility-minus-road contrast uses 10,000 paired-date resamples. Structural nulls use 999 draws; the Chengdu null uses the same seed range and pool rule, checked against a Porto control re-run to machine precision. Chengdu's three dates allow only six date permutations, so its pairing-null result is descriptive. Uncertainty follows each archived experiment's estimator, so the Chengdu equal-capacity interval resamples its four held-out dates.

Data availability

Porto and San Francisco use the cited public datasets; Chengdu uses the archived DiDi ride-hailing sample [29]. The exact Chengdu release identifier and download date were not recorded; the archived aligned field used here has SHA-256 `c9330b54e8fbcf589e8b63d480b4db877b7d48291df941fb1348662a9d`. Road data derive from OpenStreetMap and retain the OpenStreetMap contributors attribution. Frozen splits and observation regimes are tabulated in the supplementary material. The public export contains no individual-level trajectory records; redistribution status and any external-data requirements for derived fields are recorded in its data manifest.

Code availability

The analysis repository separates figure rebuilding from frozen products, core-score recomputation and end-to-end raw-trajectory reconstruction. The matched-capacity modules, phase-sensitive fre-

quency metrics, anisotropy rotation null and figure generators are included. The matched-capacity sweep still requires external aligned fields, date-freeze files and GIS rasters, which are listed with hashes in the data manifest; no raw trajectories are included. Large prediction files and original training artifacts that are not in the export are identified rather than represented as included. Execution requirements and checksums are documented in the repository manifest. Public repository location: https://github.com/Wangwangw233/paper-urbanflow.

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